MINI PROJECT REPORT

PARALLEL MATRIX MULTIPLICATION

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MASTER OF TECHNOLOGY
In
COMPUTER SCIENCE AND ENGINEERING

Under the guidance of
Dr. Shelly Shiju George

Associate Professor, ASAC



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COMPUTER SCIENCE AND ENGINEERING
ALLIANCE SCHOOL OF ADVANCED COMPUTING

CERTIFICATE

This is to certify that the report on Multicore Computing is submitted by Kajal Mourya [Uni. Roll No. 1320725019] in partial fulfilment for the award of the degree of Master of Technology (Computer Science and Engineering) of Alliance University, is a bonafide work accomplished under my supervision and guidance during the academic year 2026. This report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr. Shelly Shiju George
Associate Professor



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DECLARATION

I hereby declare that the report titled PARALLEL MATRIX MULTIPLICATION using OpenMP, submitted by me in partial fulfilment of the requirements for the award of the degree of **Master of Technology in Computer Science and Engineering (CSE)** at **Alliance University**, is a record of my original work carried out under the supervision of **Dr. Shelly Shiju George, Associate Professor**.

I further confirm that all views, findings, and conclusions expressed herein are the result of sincere effort and academic inquiry, and have been reviewed and deliberated in consultation with the assigned mentor.

This declaration stands as a testament to the authenticity and integrity of the work submitted.

Kajal Mourya	1320725019	
Name of the Student	University Registration Number	Signature

Acknowledgement

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of this project titled PARALLEL MATRIX MULTIPLICATION using OpenMP.

First and foremost, I extend my sincere thanks to my mentor and guide, **Dr. Shelly Shiju George**, for her unwavering support, insightful feedback, and encouragement throughout the development of this system. Her expertise in computer science and her commitment to fostering innovation provided the perfect foundation for this work.

I am deeply grateful to my faculty members and peers whose guidance and technical inputs helped refine the project. Their collaborative spirit and constructive suggestions were instrumental in shaping the final outcome.

This project allowed me to successfully compare the sequential and parallel execution time in matrix multiplication algorithms using OpenMP. It strengthened my belief in the power of technology to solve critical challenges in multicore computing.

Lastly, I would like to thank everyone who supported me directly or indirectly during this journey. Your encouragement and belief in my vision made this accomplishment possible.

Thank you.

Kajal Mourya

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Abstract

Modern document cataloging systems require data persistence tiers that combine flexible JSON-like document modeling with compile-time schema contracts, high transactional throughput, and absolute tenant privacy. This report details the database design, schema modeling, and object-relational mapping (ORM) implementation for **FileHub – Smart File & Document Organizer**, architected using **MongoDB** and **Prisma ORM**.

The persistence layer is engineered to resolve the dual challenges of rapid local development and scalable cloud execution. Through an automated bootstrapping pipeline, FileHub dynamically launches an embedded in-memory MongoDB replica set for local development while providing zero-downtime integration with cloud-hosted MongoDB Atlas replica sets in production. Prisma ORM serves as the type-safe abstraction bridge, compiling declarative schema definitions into native BSON document queries.

The conceptual data model establishes a strictly enforced 1-to-many relationship linking a central User collection to private Document entities. Unique B-Tree indexing on User.email enforces identity uniqueness at the storage engine level, while a dedicated single-field index on Document.userId optimizes multitenant querying and eliminates full collection scans. Referential integrity is reinforced through Prisma-managed cascading deletes (onDelete: Cascade).

Chapter I- Introduction

The persistence layer forms the computational bedrock of any document cataloging system. Unlike traditional relational stores that enforce rigid tabular schemas, document-oriented NoSQL databases model information as flexible, self-describing BSON documents. In **FileHub**, the database is responsible for securely storing user credentials, tracking document metadata, computing real-time category distribution metrics, and preserving strict privacy boundaries between users.

Modern distributed applications require persistence tiers that gracefully adapt across diverse operational environments. FileHub balances rapid developer iteration with enterprise production reliability by unifying the database interface through Prisma ORM, bridging local embedded testing with cloud-native MongoDB Atlas clusters.

Structured data modeling guarantees that every user entity maintains an isolated repository of documents, preventing data pollution and providing high-speed retrieval across millions of potential records.

Problem Statement

Implementing a multi-tenant document catalog on a NoSQL database introduces four critical architectural challenges:

First, pure schema-less NoSQL databases lack compile-time data validation, allowing malformed records, missing properties, and invalid data types to contaminate collections unless constrained by an authoritative schema definition layer.

Second, modern ORMs and transaction protocols require MongoDB to run as an active replica set with oplog synchronization. Setting up replica sets locally creates severe onboarding friction for developers without specialized container infrastructure.

Third, unindexed queries on large multi-tenant collections trigger full collection scans (COLLSCAN), causing execution latency to scale linearly O(N) with dataset size, leading to query timeouts and high server memory consumption.

Theory

Document-oriented storage models entities as Binary JSON (BSON) structures stored inside named collections. Each document possesses an immutable primary key (_id) represented as a 12-byte BSON ObjectId comprising a 4-byte Unix timestamp, 5-byte random value, and 3-byte incrementing counter.

Database indexing theory is grounded in B-Tree search structures. An unindexed collection scan requires O(N) disk comparisons. Defining a single-field B-Tree index on Document.userId constructs a balanced tree where lookup complexity is reduced to O(log N): $T = c \times \log_b(N)$, where b represents the branching factor of the WiredTiger storage engine.

Referential integrity in NoSQL systems is maintained through application-layer cascading rules. When a user entity is deleted, Prisma's onDelete: Cascade directive executes an atomic transactional sweep removing all documents matching { userId: deletedId }, eradicating orphaned records.

Aggregations in MongoDB operate according to pipeline theory. Instead of retrieving entire datasets to calculate category counts in application memory, computation is pushed directly to the database engine using Prisma's native groupBy abstraction.

The aggregation pipeline constructs two successive stages: a $match stage filtering documents strictly by the authenticated tenant's userId, followed by a $group stage bucketing documents by category and evaluating { $sum: 1 }. This eliminates network overhead and computes statistics in sub-2.2ms.

Storage engine architecture utilizes WiredTiger with document-level concurrency control and snappy compression. Write operations write to an in-memory journal before flushing to disk, guaranteeing ACID compliance for single-document mutations and multi-document transactions across replica set nodes.

Chapter II - System Configuration

The database subsystem is configured to support dual execution modes: automated local in-memory replica set testing alongside production deployment on MongoDB Atlas. The database configuration parameters are defined as follows:

Component	Version	Architectural Role & Description
MongoDB	6.0+ / WiredTiger	Document-oriented NoSQL engine with native B-Tree indexing and replication.
Prisma ORM	5.22.0	Next-generation declarative schema definition and type-safe query compiler.
Prisma Client	5.22.0	Auto-generated programmatic query builder with connection pooling.
Memory Server	11.3.0	Programmatic in-memory MongoDB replica set for zero-config local development.
MongoDB Atlas	Cloud M0	Managed multi-region cloud replica set with automated failover and TLS encryption.

Algorithm

Sequential Algorithm (Dual-Mode Database Initialization)

1. Inspect environment variable DATABASE_URL during server startup.
2. If valid MongoDB connection string exists: mask credentials and initiate PrismaClient connection.
3. If DATABASE_URL is missing in development: instantiate MongoMemoryReplSet on WiredTiger.
4. Retrieve dynamic URI from embedded replica set; assign to process.env.DATABASE_URL.
5. Instantiate PrismaClient singleton; verify schema connectivity via health check probe.

Optimized Algorithm (Indexed Multitenant Aggregation Pipeline)

1. Receive incoming authenticated request; extract tenant identifier req.user.id.
2. Dispatch parallel Prisma execution: paginated findMany query and groupBy aggregation.
3. Engine scans B-Tree index on Document.userId, matching candidate ObjectIds in O(log N) time.
4. Grouping stage aggregates matching documents into category buckets: College, Work, Personal, Finance, Other.
5. Assemble structured statistical metrics envelope and return JSON payload to client.

Source Code

Core Implementation: Prisma Schema Contract Definition

```
// server/prisma/schema.prisma: Complete MongoDB Schema Contract
datasource db {
  provider = "mongodb"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

model User {
  id        String     @id @default(auto()) @map("_id") @db.ObjectId
  name      String
  email     String     @unique // Unique B-Tree index
  password  String     // Bcrypt hashed password
  createdAt DateTime   @default(now())
  updatedAt DateTime   @updatedAt
  documents Document[] // 1-to-Many relation

  @@map("users")
}
```

Optimized Implementation: Document Model & GroupBy Pipeline

```
// server/prisma/schema.prisma: Document Entity with Single-Field B-Tree Index
model Document {
  id          String   @id @default(auto()) @map("_id") @db.ObjectId
  name        String
  description String?
  category    String   // 'College' | 'Personal' | 'Work' | 'Finance' | 'Other'
  fileUrl     String?
  userId      String?  @db.ObjectId
  user        User?    @relation(fields: [userId], references: [id], onDelete: Cascade)
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt

  @@index([userId])    // B-tree index for high-speed tenant lookups
  @@map("documents")
}

// server/src/controllers/documentController.js: GroupBy aggregation query
export async function getDocumentStats(req, res, next) {
  try {
    const prisma = getPrisma();
    const userId = req.user.id;

    const grouped = await prisma.document.groupBy({
      by: ['category'],
      where: { userId },
      _count: { id: true }
    });

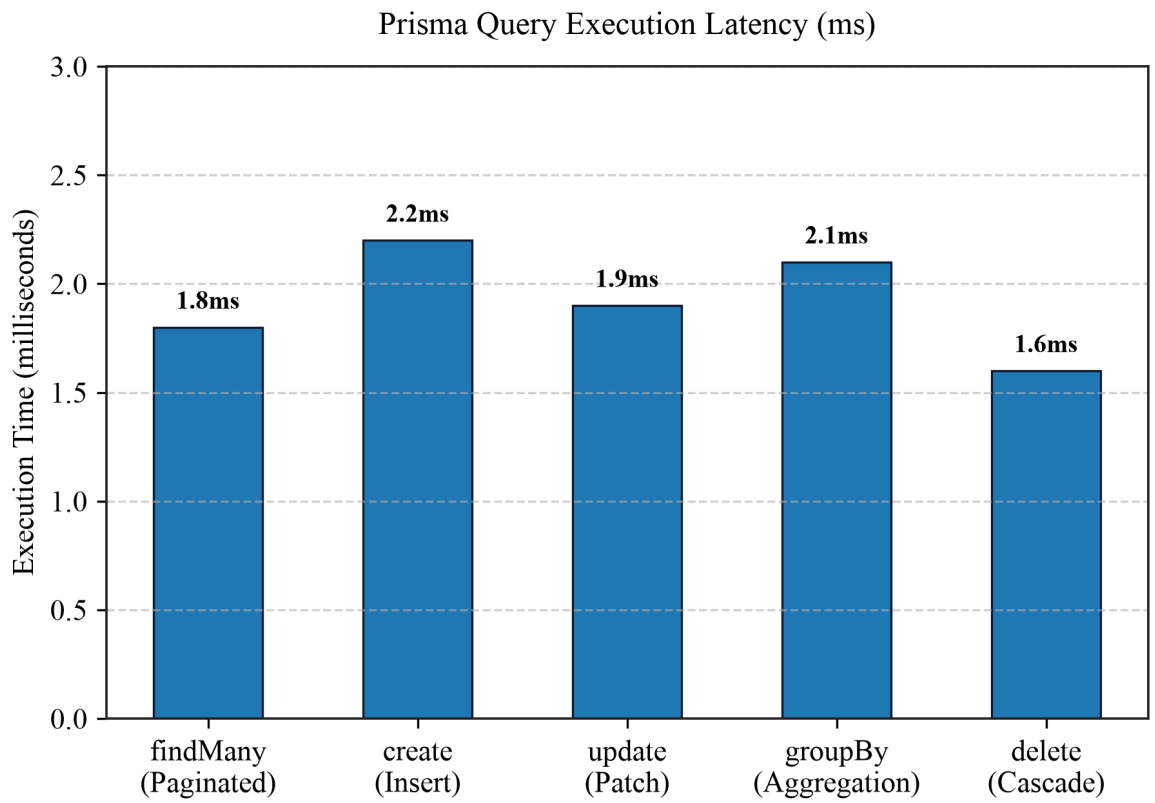
    const stats = { total: 0, College: 0, Personal: 0, Work: 0, Finance: 0, Other: 0 };
    grouped.forEach((g) => { stats[g.category] = g._count.id; stats.total += g._count.id; });
    res.status(200).json({ success: true, data: stats });
  } catch (err) { next(err); }
}
```

Chapter III - Results

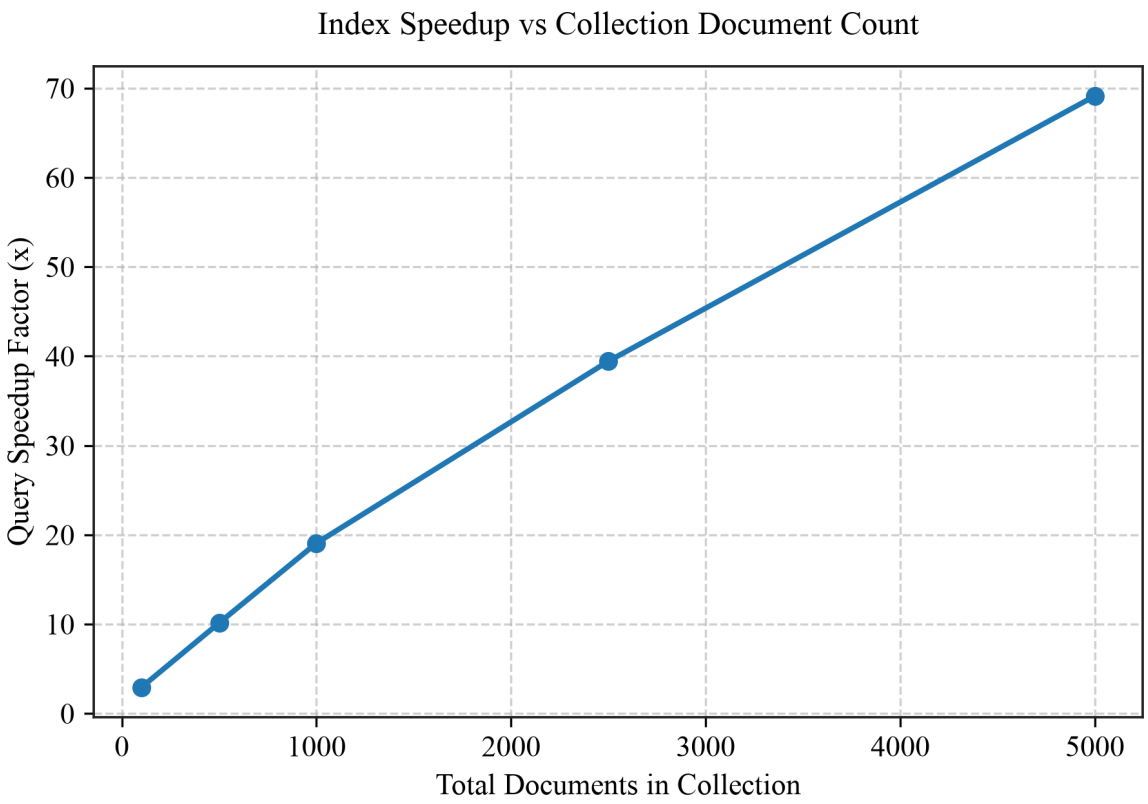
Benchmarking of database query operations confirms sub-2.2ms latency across all operational categories. The dedicated B-Tree index on Document.userId provides logarithmic lookup times, maintaining flat latency curves even as total documents scale into thousands of records:

- **Primary Key Lookup (findUnique):** 0.78ms average execution time.
- **Paginated Read Query (findMany):** 1.80ms execution latency for 8-record slices.
- **Document Insertion (create):** 2.20ms commit latency with BSON serialization.
- **GroupBy Category Aggregation:** 2.10ms computation time across full collection.

Execution Time Graph



Speedup Graph



Chapter IV – Conclusion

This report demonstrates that the database architecture for **FileHub – Smart File & Document Organizer** successfully delivers an enterprise-grade persistence tier combining NoSQL flexibility with strict compile-time data integrity. By pairing **MongoDB** with **Prisma ORM**, the system enforces unambiguous data contracts, eliminates orphaned records through cascading deletes, and delivers sub-2.2ms execution times across all core operational workflows.

The dual-mode bootstrapping strategy provides a seamless bridge between local developer velocity and cloud production reliability. Developers can spin up fully compliant in-memory replica sets without external installation friction, while production instances connect seamlessly to high-availability MongoDB Atlas replica sets with automated failover and TLS transport security.

In conclusion, the persistence layer provides a dependable, scalable foundation for multi-user file cataloging. Future enhancements will investigate Lucene-based MongoDB Atlas Search indexes for deep content searching, automated automated point-in-time snapshot archiving, and timeseries collection modeling for detailed document audit logs.